

**Time:** 6 minutes**Outcome:** Calculate the real exchange rate and use PPP as a long-run benchmark.**Difficulty:** ●●○ Intermediate

Real Exchange Rates and Purchasing Power

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THE CORE IDEA

Let e be foreign currency per domestic currency, P the domestic price level, and P^* the foreign price level. The course convention is $\epsilon = e \times P / P^*$. A higher epsilon is a real appreciation: domestic goods are relatively more expensive. Purchasing-power parity suggests epsilon tends toward 1 when comparable traded goods can be arbitrated, but transport costs, trade barriers, nontraded goods, and market frictions prevent exact short-run equality.

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HOW TO RECOGNIZE IT

- Nominal e compares currencies; real epsilon compares basket prices after conversion.
- A higher e or P raises epsilon; a higher P^* lowers epsilon, other things equal.
- Real appreciation tends to reduce exports and raise imports.
- Relative PPP links currency changes to inflation differentials over longer horizons.

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WATCH OUT

Do not invert the formula by importing another textbook's quotation. Write e 's units first. Here e is foreign currency per domestic currency, so $\epsilon = eP/P^*$.

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WORKED EXAMPLE: CONVERTING TWO BASKETS

KEY RELATIONSHIPS

- e : foreign currency / domestic currency
- $\epsilon = e \times P / P^*$
- Higher epsilon = real appreciation
- PPP benchmark: epsilon near 1
- Approx.: % change epsilon = % e + domestic inflation - foreign inflation

A U.S. basket costs \$50, a comparable foreign basket costs 40 foreign units, and $e = 0.80$ foreign units per dollar. Then $\epsilon = 0.80 \times 50 / 40 = 1.00$. If the U.S. price rises to \$55 with e and P^* fixed, epsilon becomes 1.10: a 10% real appreciation.

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CHECK YOURSELF

With e and P fixed, foreign prices rise. Does epsilon rise or fall under the course formula?

**READY?****Return to the game and master this concept.**